

Gift A. Nwatuzie

Ph.D. Candidate (ABD) in Computer Science
Adjunct Faculty

+1 (206) 291-3094
gnwatuzi@kent.edu
gifttest.github.io

Department of Computer Science
Kent State University
Kent, Ohio, USA

RESEARCH INTERESTS

Machine Learning; Trustworthy AI; Privacy-Preserving Machine Learning; Federated Learning; Explainable and Fair AI; Multimodal Learning; Cybersecurity; Secure Distributed Systems; Software Engineering for AI Systems.

EDUCATION

Kent State University
Ph.D. in Computer Science

Kent, Ohio, USA
Spring 2027 (Expected)

University of East London
M.Sc. in Computer Systems Engineering

London, UK
August 2019

University of East London
B.Sc. in Software Engineering

London, UK
June 2018

TEACHING EXPERIENCE

Kent State University
Adjunct Faculty

January 2026–Present

Kent State University
Graduate Teaching Assistant

August 2022–December 2025

Kent State University
Graduate Hourly Employee Coordinator

January 2024–December 2025

COURSES TAUGHT AND SUPPORTED

CS 46101/56101-020 Design and Analysis of Algorithms
CS 13001 Computer Science I: Programming and Problem Solving
CS 13012 Object-Oriented Programming
CS 13011 Procedural Programming

Instructor.
Teaching Assistant.
Teaching Assistant.
Teaching Assistant.

INDUSTRY EXPERIENCE

Coding Expert Bootcamp (Global Fellowship)
Software Engineering Fellow

March 2021–September 2022

Techvision Volunteer
Lead Software Architect & Ethical AI Advisor

January 2021–December 2022

Trendsville / Pack Media Ltd
Software Engineer

January 2020–February 2021

Tech Nova Solutions
Software Engineer

May 2019–July 2026

Civil Service Commission
Software Engineer

January 2016–December 2019

RESEARCH MANUSCRIPTS

Paper in Preparation

- Gift Nwatuze and Hassan Peyravi, “ZKPF: Recursive Zero-Knowledge Proofs for Verifiable and Privacy-Preserving Provenance in Multi-Modal Sensor-AI Pipelines,” to be submitted to *Sensors*, MDPI, October 2026.

Papers under Revised Review

- Gift Nwatuze and Hassan Peyravi, “FairAudit: A Causal Explanation Framework for Diagnosing and Mitigating Intersectional Bias,” submitted to *AI and Ethics*, Springer, May 2026.
- Gift Nwatuze and Hassan Peyravi, “Federated Learning for Healthcare Secure Data Sharing,” submitted to *cybersecurity journal*, Springer, July 2024.

Paper under Review

- Gift Nwatuze and Hassan Peyravi, “Quantum-Resilient Federated Learning with Trust-Aware Secure Aggregation for Responsible Multi-Modal Intelligence,” submitted to *ACM (Transactions on Multimedia Computing, Communications, and Applications)*, August 2026.

Book Chapter Accepted

- Gift Nwatuze and Hassan Peyravi, “Adaptive Quantum–Classical Intrusion Detection Framework (AQCID),” *Large Language Models: AI Tools, NLP and Use Cases*, Springer, August 2026.

JOURNAL PUBLICATIONS

1. Gift Nwatuze and Hassan Peyravi, “A Framework for Integration of Machine Vision with IoT Sensing,” *Sensors*, 2025.
2. Gift Nwatuze and Hassan Peyravi, “Quantum-Enhanced Intrusion Detection: A Novel Hybrid Approach Using Quantum Deep Learning,” 23rd International Conference on Scientific Computing (CSC), Las Vegas, USA, CCIS, Springer, December 2025.
3. Gift Nwatuze and Hassan Peyravi, “Deep Learning-Based Data Recovery System for Network Monitoring (DL-DRSNM),” International Conference on Computational Science and Computational Intelligence (CSCI), Las Vegas, USA, CCIS, Springer, July 2024.
4. Gift Nwatuze, “Comparative Performance and Optimization Strategies for Cloud-Native Architectures: A Focus on Scalability, Cost, and Resource Utilization,” *International Journal of Information Engineering and Electronic Business*, 2026.
5. Gift Nwatuze, Lawrence Anebi Enyejo, and Chima Umeaku, “Enhancing Cloud Data Security using a Hybrid Encryption Framework Integrating AES, DES, and RC6 with File Splitting and Steganographic Key Management,” *International Journal of Innovative Science and Research Technology*, 2025.
6. Gift Nwatuze, “Enhancing Security and Resilience in Cyber-Physical Systems through AI-Driven Threat Detection, Formal Verification, and Blockchain Technologies,” *International Journal on Science and Technology*, 2025.
7. Gift Nwatuze, Onu Mathew Ijiga, Idoko Peter Idoko, Lawrence Anebi Enyejo, and Emmanuel Olache Ali, “Design and Evaluation of a User-Centric Cryptographic Model Leveraging Hybrid Algorithms for Secure Cloud Storage and Data Integrity,” *American Journal of Innovation in Science and Engineering*, 2025.
8. Gift Nwatuze, “Digital Twin Technology in Software Engineering: Simulation and Testing in Virtual Environments,” *World Journal of Engineering Research*, 2025.
9. Gift Nwatuze, “Human-Centered Software Engineering: Enhancing Developer Productivity and Well-Being,” *World Journal of Engineering Research*, 2025.
10. Gift Nwatuze, “Blockchain-Based Secure Software Engineering Practices,” *International Journal for Research in Applied Science and Engineering Technology*, 2025.
11. Gift Nwatuze, “Automated Software Refactoring Utilizing Machine Learning for Extensive Systems,” *International Journal for Research in Applied Science and Engineering Technology*, 2025.
12. Gift Nwatuze, “Adaptive Federated Learning for Secure and Efficient Edge Intelligence in Mobile Computing,” *World Journal of Engineering Research and Technology*, 2022.
13. Gift Nwatuze, “Blockchain-Enabled Federated Learning for Trustworthy Multi-Agent Systems,” *World Journal of Engineering Research and Technology*, 2021.
14. Gift Nwatuze, “Enhanced Intrusion Detection System with Machine Learning Models and Class Imbalance Optimization,” *World Journal of Engineering Research and Technology*, 2019.
15. Gift Nwatuze and Kwok Wai Jing, “Android Device File Encryption and Decryption System,” *International Journal of*

Information System and Engineering, 2019.

RESEARCH PRESENTATIONS

- “Quantum-Enhanced Intrusion Detection: A Novel Hybrid Approach Using Quantum Deep Learning,” CSC, December 2025.
- “Deep Learning-Based Data Recovery System for Network Monitoring (DL-DRSNM),” CSCI, July 2024.
- “Android Device File Encryption and Decryption System,” IJISE, April 2019.

CERTIFICATIONS

- CompTIA Security certification.
- Cybersecurity Threats and Defense certification.
- Cybersecurity Operations and Controls certification.
- Information Security certification.

PROFESSIONAL AFFILIATIONS

- Institute of Electrical and Electronics Engineers (IEEE).
- IEEE Women in Engineering (IEEE WIE).
- Association for Computing Machinery (ACM).
- Sigma Xi, Associate Member.
- Nigerian Institution of Professional Engineering and Scientists (NIPES), Fellow.

ACADEMIC HONORS

- NSF Travel Award, IEEE PerCom 2025 Conference, Washington, D.C., March 17–21, 2025.
- Certificate of Best Diplomats, Master Diplomacy, Leadership and Negotiations, USA, February 2025.
- Award for Excellence in Innovation, Rivers State, Port Harcourt, Nigeria, 2020.
- Merit Award — Exceptional Woman in Technology, Rivers State, Nigeria, 2019.
- STEM Outstanding Achievement Award, Rivers State, Nigeria, 2018.
- Dean’s Award for Academic Excellence, University of East London, UK, 2015.

JOURNAL REVIEW ACTIVITIES

- Cogent Arts and Humanities.
- Next Nanotechnology, Elsevier.
- International Journal of Information Security (IJIS).
- Frontiers in Computer Science and Information Technology (FCSIT).
- International Journal of Artificial Intelligence and Deep Learning (IJAILD).
- International Journal of Artificial Intelligence and Machine Learning (IJAIML).